

A density-characteristic temperature T_n from the logarithmic slope of f for the `isc=2` self-collision background

FP2D_QLRF_NL — design note

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This note gives a complete, self-consistent definition of a density-characteristic temperature T_n that can be used in place of the energy-weighted T_{eff} as the background temperature of the `isc=2` self-collision operator, with the velocity convention pinned down so it is a drop-in replacement.

1 Convention (what the operator actually uses)

`cblln` builds its background Maxwellian as the standard Maxwellian

$$f_M \propto \exp\left(-\frac{v^2}{2v_{\text{th}}^2}\right), \quad v_{\text{th}} = 9.79 \times 10^3 \sqrt{T[\text{eV}]/A} = \sqrt{T/m}, \quad m = A m_p. \quad (1)$$

(In `cblln`, `arg = v1/(\sqrt{2} vth)` and `arg2 = v^2/(2 vth^2)` — i.e. the Maxwellian at temperature T .) A temperature is therefore turned into a background through $v_{\text{th}}^2 = T/m$. We define T_n so that it plugs straight into that same mapping; then it is a drop-in replacement for T_{eff} in the `isc=2` branch.

2 Building block: the local log-slope temperature

For that Maxwellian, $\ln f = \text{const} - v^2/2v_{\text{th}}^2$, so the radial logarithmic slope is

$$\frac{\partial \ln f}{\partial v} = -\frac{v}{v_{\text{th}}^2} \implies v_{\text{th}}^2(\mathbf{v}) \equiv \frac{-v}{\partial_v \ln f} = \frac{-v^2}{\mathbf{v} \cdot \nabla_v \ln f}, \quad T_{\text{loc}}(\mathbf{v}) = m v_{\text{th}}^2(\mathbf{v}). \quad (2)$$

This is the pointwise “temperature seen in the slope of f ”. It is exact (constant, equal to T) for a Maxwellian and varies across velocity space for a non-Maxwellian f .

3 The scalar T_n (density-weighted log-slope)

Average the *inverse* local thermal speed, weighted by the distribution itself (so the dense bulk dominates):

$$\boxed{\frac{1}{v_{\text{th},n}^2} \equiv \frac{1}{n} \int f \frac{-\mathbf{v} \cdot \nabla_v \ln f}{v^2} d^3v = \frac{1}{n} \int \frac{-\mathbf{v} \cdot \nabla_v f}{v^2} d^3v}, \quad T_n = m v_{\text{th},n}^2. \quad (3)$$

The second form (no $1/f$) is the one to implement. Integrating it by parts, using $\nabla_v \cdot (\mathbf{v}/v^2) = 1/v^2$ (boundary terms vanish), gives a remarkably clean closed form:

$$\boxed{\frac{1}{v_{\text{th},n}^2} = \left\langle \frac{1}{v^2} \right\rangle \equiv \frac{1}{n} \int \frac{f}{v^2} d^3v \implies T_n = \frac{m}{\langle v^{-2} \rangle}} \quad (4)$$

So the “log-slope temperature” *is exactly the harmonic mean of v^2* . Both forms are mathematically identical; the $\langle v^{-2} \rangle$ form is preferable numerically (no derivatives of f).

Why this is the right “density-characteristic” temperature.

- It reduces to T for a Maxwellian: for $f \propto \exp(-v^2/2v_{\text{th}}^2)$ one finds $\langle v^{-2} \rangle = 1/v_{\text{th}}^2$, hence $T_n = T$.
- $\langle v^{-2} \rangle$ is dominated by *small* v (the dense, cold core), so T_n tracks the bulk and largely ignores the hot perpendicular tail. Contrast this with $T_{\text{eff}} \propto \langle v^2 \rangle$, which is tail-weighted. For the RF case this gives $T_n \approx \text{bulk} (\sim 12 \text{ keV})$ while $T_{\text{eff}} \approx 22 \text{ keV}$ — exactly the gap that drives the `isc=2` pathology.

4 Code recipe (2D cylindrical, existing machinery)

$$\langle v^{-2} \rangle = \frac{\sum_{i,j} \frac{f_{ij}}{v_{\perp i}^2 + v_{\parallel j}^2} J_{ij}}{\sum_{i,j} f_{ij} J_{ij}}, \quad T_n[\text{eV}] = \frac{A}{(9.79 \times 10^3)^2 \langle v^{-2} \rangle}, \quad (5)$$

where J_{ij} is the existing `jacob` ($2\pi v_{\perp} \times$ grid weights). Concretely, mirroring `time_energy`:

```
! density-characteristic (log-slope) temperature; code convention vth^2 = T/m
do iv = 1, nperp
  do ip = 1, npar
    v2 = vperp(iv)**2 + vpar(ip)**2
    fint(iv,ip) = f(iv,ip) / max(v2, v2_floor) * jacob(iv,ip)    ! integrand <1/v^2>
  end do
end do
call ncint_2d(fint, mom_invv2)          ! = n * <1/v^2>
inv_v2_avg = mom_invv2 / dens          ! <1/v^2>
Tn_eV = aa / (9.79d3**2 * inv_v2_avg)  ! drop-in for ta_eV/1e3 in the isc=2 branch
```

- **Origin guard:** the integrand $\propto f/v^2$ has only a mild, integrable singularity at $\mathbf{v} = 0$; since `vperp(1)  $\approx 394 \text{ ms}^{-1} > 0$` there is no exact zero, but keep a small `v2_floor` (e.g. `vperp(1)**2`) for safety.
- To use it, replace the `isc=2` line `vteff_t = 9.79d3*SQR(ta_eV/aa)` by `vteff_t = 9.79d3*SQR(Tn_eV/aa)` (and use `Tn_eV` for the self-collision Coulomb log). Keep writing both `Teff_vs_time` and a new `Tn_vs_time` so that the gap T_{eff}/T_n is visible.

Remark: a related convention point

The diagnostic SC-Maxwellian output (`fsc_maxw` in `TimeFP_7pt.f90`, and `stix_maxw_plot.py`) originally used $\exp(-v^2/v_{\text{th}}^2)$ with prefactor $1/(\pi^{3/2}v_{\text{th}}^3)$ — too narrow by $\sqrt{2}$ and mis-normalised relative to the $\exp(-v^2/2v_{\text{th}}^2)$, $1/((2\pi)^{3/2}v_{\text{th}}^3)$ background that `cbilin` actually uses. This has since been corrected so the diagnostic matches the operator. It does not affect the T_n definition above, which is stated directly in the `cbilin` convention.